

Molecular characterisation and recent evolution of myxoma virus in Spain

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Abstract Myxoma virus and the European rabbit are one of the best-studied examples of coevolution of pathogen virulence and host resistance. Since the introduction of the virus in Spain in 1953, a decrease in its virulence has been observed; however, most strains are still considered highly virulent. To determine whether this attenuation is due to molecular differences, and to characterise the field strains in Spain and the genetic changes that have occurred since the introduction of the virus, we analysed 7,741 bp in 97 virus samples from 12 localities. We found an extremely low genetic variability and an absence of a geographic structure. We defined 35 haplotypes, none of which were identical to the original Lausanne strain. Three genetic groups were determined and were found to occur at different frequencies in different locations. Overall, virus evolution deviated from neutrality and did not conform to a strict molecular clock, probably due to the existence of a strong selective pressure that acts differently across the viral genome.

Introduction

Myxoma virus (MV) is a member of the genus *Leporipoxvirus*, family Poxviridae [23]. Its natural host is the American rabbit of the genus *Sylvilagus*, which, upon MV infection, suffers from benign localised skin fibromas. However, in the European rabbit (*Oryctolagus cuniculus*), MV is the agent that causes myxomatosis, a severe disease that produces a systemic immunosuppression that is deadly in most cases [24].

The discovery of MV's high specificity for *O. cuniculus*, and of its relatively easy mechanical transmission mediated by arthropods, mainly mosquitoes and fleas, led researchers to consider the use of this virus as a control agent for the overpopulation of rabbits in Australia. The release of the MV was legally authorised in 1950 [58], and in just 2 or 3 years, rabbit populations were reduced by 95% [70]. Shortly thereafter, in 1952, the MV Lausanne strain was illegally introduced in France. This strain resulted in a much more virulent form of MV than the SLS strain that had been introduced previously in Australia [22]. The disease rapidly spread across Europe, causing high mortality, and in 1953, the first case of myxomatosis was reported in northern Spain [49].

After the first epizooties, the incidence of myxomatosis in Australia and Europe started to decrease as a result of the emergence of less pathogenic MV strains and rabbits genetically resistant to the disease [22, 36]. Consequently, the more virulent MV strains progressively disappeared, whereas those that had moderate virulence became more widespread [25, 44]. At the same time, low-virulent strains allowed the rabbits with an innate resistance to the virus to survive the infection [43, 45]. As a consequence of this, there currently exists an equilibrium between the MV's virulence and the host's resistance [36]. This equilibrium is one of the best-studied examples of coevolution between a host and its pathogen [3, 7,

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8, 43], as is the immunosuppressive mechanisms used by MV in both its virulent and non-pathogenic forms [33, 50, 68].

The complete MV genome has been sequenced, and the putative functions of most of its 156 reading frames have been deduced [9]; however, the molecular causes that account for the attenuation of the virus are not fully understood [48]. Most of the molecular epidemiology studies on this virus have used RFLPs to investigate the differentiation between the different MV types. In all cases, low divergences were reported among strains from different regions [41, 61] or among those sampled at different points in time in Australia. In addition to the low variation found between Australian samples and the originally introduced strain, no association has been found between the RFLP profiles and their virulence [62]. In Spain, previous studies dealing with the virulence of MV have identified a non-pathogenic field strain that could be used as a vaccine [6]. The genome of this non-pathogenic strain is 99.95% similar to that of its ancestor, the Lausanne strain. Only four genes have been disrupted and could be responsible for the attenuation of the MV [48]. However, in Europe, the genetic variability of MV in the wild has never been studied.

In Spain, 75% of the field strains still show a high or very high virulence, as does the Lausanne strain, which was originally introduced from France [6, 21, 49]. Considering the decrease in rabbit mortality caused by myxomatosis, we could assume that the virus or the host, or both, should have experienced a series of genetic changes since the introduction of this virus 56 years ago. We would consequently expect (1) to observe different MV strains, which should show genetic variation among them and with the original Lausanne strain. Additionally, (2) the genetic changes that confer an advantage to the survival of the MV should be positively selected and consequently, (3) the frequency of these strains should increase [28].

Here, we report the results of our study, which describes the genetic variability of MV in the wild, and try to infer an evolutionary model that could predict this variability. An initial step to understand the evolution of myxomatosis and to determine how it affects different geographical regions is to identify the MV field strains. Thus, the specific aim was to (1) molecularly characterise the MV field strains in Spain, (2) study their population genetic structure and (3) determine the intrapopulation evolution of the MV since its introduction to Spain in 1953.

Materials and methods

Sample collection

We detected the presence of MV in rabbits from 12 locations of the Iberian Peninsula and Balearic Islands,

collected from 2004 to 2007: (1) Mallorca, (2) Galicia, (3) Madrid1, (4) Madrid2, (5) Toledo1, (6) Toledo2, (7) Toledo3, (8) Ciudad Real, (9) Albacete, (10) Badajoz, (11) Jaén, (12) Sevilla (Fig. 1). Animals were either captured alive or found dead with symptoms of myxomatosis. Samples from the conjunctival membrane of the eye were collected with sterile Dacron[®] swabs and stored at 4°C until DNA extraction.

DNA extraction and amplification

Viral DNA was isolated using a QIAamp[®] DNA Mini kit (QIAGEN) following the buccal swab protocol, and the DNA was diluted to a final volume of 100 µl.

We amplified by PCR 7,681 bp in seven fragments distributed across the whole MV genome. Primers were designed according to the complete genome sequence of the MV Lausanne strain [9] and included reading frames with different functions such as immunoregulation, host specificity, structural, housekeeping, and other unknown functions (Table 1).

First, a single fragment (M077–M079) was amplified for all the samples, and those that resulted in a positive amplification were subsequently amplified to obtain the other six fragments. PCR reactions were prepared in a final volume of 25 µl containing 200 µM of dNTPs, 0.2 µM of each primer, 1 µg/µl of BSA, 1U of *Taq* polymerase (Eppendorf), 1× PCR reaction buffer (500 mM KCl, 100 mM Tris–HCl, pH 8.3, 15 mM Mg²⁺) and 2 µl of the DNA extract. The PCR programme was as follows: 4 min of denaturation at 94°C, 35 cycles of 1 min at 94°C, 1 min at 55°C and 1 min 30 s at 72°C, and a final elongation step of 10 min at 72°C. All of the reactions included positive and negative controls.

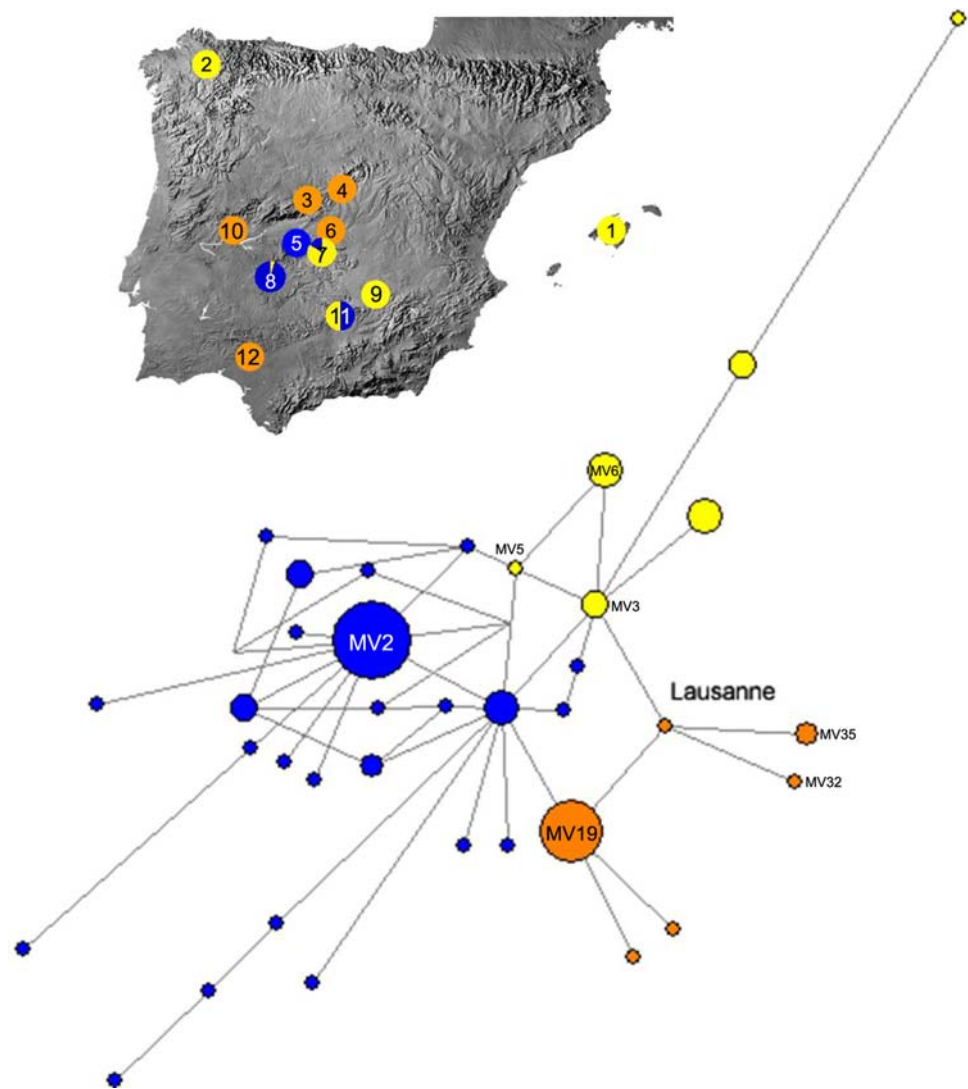
PCR products were purified with ExoSAP-IT (GE Healthcare) and sequenced in an ABI3730 automated sequencer using a BigDye Terminator v3.1 kit (Applied Biosystems) and the same PCR primers.

The chromatograms were checked and aligned by eye with respect to the homologous sequences of the Lausanne strain (NC_001132). Nucleotide sequences were translated into amino acids to construct three different alignments. The first alignment contained all of the data, while the second only comprised the coding regions, excluding the stop and incomplete codons, and the third one only encompassed the non-coding regions.

Polymorphism analysis

To evaluate the genetic variability of MV, we calculated for each alignment its genetic diversity (H_d), nucleotide diversity (π), number of polymorphic sites (S), average number of nucleotide differences (k), and expected heterozygosity by site (θ) in DnaSP 4.20.2 [60].

Fig. 1 Map showing the locations where MV samples were analysed and the median-joining network created for all of the MV haplotypes found in Spain. *Numbers* on the map correspond to the locations mentioned in the text, and the *colours* on the map and in the network represent the genetic groups that were inferred in BAPS for MV. Group A grey, Group B dark grey, Group C light grey (colours in the online version, A orange; B blue; C yellow)



The demographic model that best fit our data was estimated using a coalescent-based Bayesian method, using the software BEAST 1.4.6 [18]. We analysed the entire data set using three coalescent models: population of constant size, exponential growth and logistic growth, assuming a strict and a relaxed molecular clock. To determine the best fit of these models to the data, we calculated their Bayes factors (BF) with 1,000 bootstrap replicates in Tracer 1.4 [56]. This calculation was done using the harmonic mean of the sampled likelihoods as an estimator of the marginal likelihood values [51]. It is thus possible to calculate the logBF from the difference between the harmonic mean and the product of the tree and the coalescent likelihood values from each model. Generally, logBF values larger than 2 are considered to significantly support a specific model [34].

Based on the best model chosen, we calculated the mutation rate for each of the three alignments in BEAST 1.4.6. To do this, we included the date on which the

samples were isolated and used the Lausanne strain to represent the first epidemics in Spain in 1953 [49].

The demographic history of MV was characterised using a mismatch distribution analysis [64], implemented in DnaSP 4.20.2. The observed distribution between inter-sample nucleotide differences for the three alignments was tested against a constant population size and a sudden-expansion model. Deviations from the sudden-expansion model were tested using the Harpending raggedness index (Hri) [31] and 1,000 replicates of coalescence simulations, to evaluate its statistical significance, in DnaSP 4.20.2.

Neutrality and positive selection tests

Deviations from the neutral evolutionary model were assessed using Tajima's D [66], Fu and Li's D^* and F^* [26], Fu's F [27] and R_2 [57]. All of these statistics were calculated in DnaSP 4.20.2, and their statistical significance was

Table 1 Primers used for the amplification of genome fragments of MV

Fragment	Primer (5'–3')	Position in the MV genome	Size (bp)	Locus type
M011–M013	11F: CCGCTACGTACATTGTAGTATG 13R: TCATCTATCTAGCGATGGAGG	13,767–14,860	1,094	IR, HK, UK
M029–M031	29F: ATTTTCTCGTACTTGATCGTGTC 31R: ATCTCTCTGAACCATATGACGTCATC	29,256–30,268	1,013	IR, HK, ST
M068–M071	68F: GGAACGGGATACTACAAGTACTT 71R: GTACGTTAATCACGTACATCGTG	66,019–67,178	1,160	HK, ST, UK, ST
M077–M079	77F: TTTTCTAACACAATAAAATCGTTCGTTGTA 79R: TGATCGTCTTTCAAAATCCACTC	75,538–76,402	865	ST, ST, HK
M089–M091	89F: CAATATCCTGGAGACTGACGCGT 91R: ACTCAGTCACGTTGGATCCGCC	89,014–89,911	898	HK, HK, UK
M128–M131	128F: CGTGTAGGGCGTTACGGTACTT 131R: TTACCCAGATCTCCGACGTGTCGG	121,871–23,055	1,185	IR, UK, UK, HK
M150–M152	150F: GAGATCAAATTTAAATCCTTAATAA 152R: TCGTCACATGGGAGTTCTATGATCTT	146,634–148,099	1,466	IR, IR, IR

IR denotes loci with immunoregulation functions or host specificity, *ST* indicates structural loci or loci related to protein morphogenesis and modification, *HK* indicates housekeeping genes and *UK* stands for loci with an unknown function [9]

calculated using 1,000 coalescence simulation replicates. Tajima's D , Fu and Li's D^* and F^* , and R_2 use information from the frequency of mutations, i.e., the number of polymorphic sites. Thus, these statistics can be useful to discriminate between growing populations and those with constant size, because we would expect the former to generate excess mutations in the genealogy's outer branches (recent mutations) and therefore a surplus of singletons [64, 66]. On the other hand, Fu's F is based on the haplotype frequency distribution relative to the nucleotide diversity [27].

To determine the existence of positive selection, we calculated the ratio between non-synonymous (d_N) and synonymous (d_S) substitutions for the whole of the coding region and for each individual codon, using three complementary methods based on maximum likelihood: SLAC (single likelihood ancestor), FEL (fixed effects likelihood) and REL (random effects likelihood) [37]. The analyses were performed online using the Datamonkey package [38]. The presence of recombination in a data set of sequences can change the evolutionary history representation of the samples into a single tree. Therefore, we first analysed our data to detect possible recombination points with the GARD [39] method in Datamonkey. We established a value of $\alpha = 0.1$ for SLAC and FEL and $BF = 10$ for REL. In all cases, the d_N/d_S ratio was calculated using the neighbour-joining (NJ) tree for the MV haplotypes.

Phylogenetic relationships and genetic structure

Unique MV haplotypes were analysed in PAUP* v4.0b10 [65] using a NJ phylogenetic method. For the NJ analysis, the

evolutionary model that best fit our data, after the Akaike criterion, was estimated in Modeltest 3.6 [53]. The support of the inferred NJ tree was assessed with 1,000 bootstrap replicates. We also represented the genealogical relationships of all the analysed samples with a haplotype network calculated from the median-joining algorithm [5] in NETWORK 4.5 (available at <http://www.fluxus-engineering.com>).

To study the genetic structure of MV in Spain, we used the Bayesian clustering method in BAPS 4.14, which calculates the most likely number of genetic groups in which the analysed population is subdivided [12]. Furthermore, most of the Bayesian clustering methods assume that all markers are independent and consequently distributed across all of the genome. When these assumptions are violated, it is erroneously assumed, according to an independence model, that the empirical data contain more information [13]. To consider this dependence of the DNA characters, we created a linkage map of the analysed loci, and we used the “clustering with linked loci” option in BAPS 4.14.

Results

We detected the presence of MV in 97 samples. The infected samples belonged to 12 localities: Mallorca ($n = 5$), Galicia ($n = 4$), Madrid1 ($n = 6$), Madrid2 ($n = 6$), Toledo1 ($n = 8$), Toledo2 ($n = 3$), Toledo3 ($n = 3$), Ciudad Real ($n = 45$), Albacete ($n = 6$), Badajoz ($n = 2$), Jaén ($n = 6$) and Sevilla ($n = 3$) (Fig. 1). For all the samples, we amplified the seven fragments, which ranged in size from 865 to 1,466 bp (Table 1), yielding a

Table 2 Genetic variability parameters for MV

	<i>n</i>	<i>bp</i>	<i>h</i>	<i>H_d</i>	<i>SD</i>	<i>S</i>	π	<i>SD</i>	<i>k</i>	θ	<i>SD</i>	Substitution rate	Lower 95% HPD	Higher 95% HPD
Entire data set	97	7,741	35	0.895	0.022	33	0.00036	0.00003	2.738	0.00083	0.00025	0.00006264	0.00003691	0.00009002
Coding regions	97	7,179	34	0.889	0.023	30	0.00038	0.00003	2.698	0.00081	0.00024	0.00006516	0.00003989	0.00009799
Non-coding regions	97	453	3	0.061	0.033	3	0.00018	0.00011	0.082	0.00129	0.00078			

n Number of samples, *bp* base pairs, *h* haplotype number, *H_d* haplotype diversity, *S* number of polymorphic sites, π nucleotide diversity, *k* average number of polymorphic sites, θ expected heterozygosity by site, *SD* standard deviation

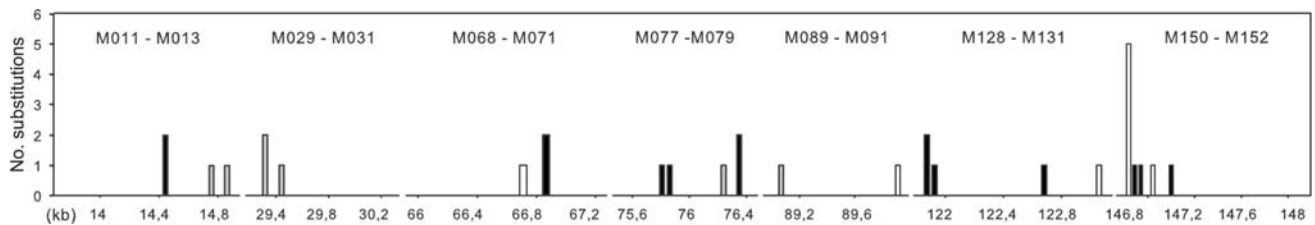


Fig. 2 Schematic representation of the analysed MV fragments and the number of polymorphic sites found. White bars represent positively selected sites, black bars indicate negatively selected sites and grey bars are representative of those sites where no selection was detected

final alignment of 7,681 bp. The alignments of the coding and non-coding regions were 7,179 and 453 bp long, respectively. Sequences were deposited in GenBank under accession numbers GQ398506–GQ398750 (Supplementary Material 1).

Overall, the genetic variability found for MV was low (Table 2). We found 35 haplotypes among the 97 MV samples ($H_d = 0.895 \pm 0.022$), with an average of $k = 2.738$ nucleotide substitutions between haplotypes. None of the haplotypes found were identical to the Lausanne strain. The most common haplotype (MV2) was found in 27 samples from two locations (Ciudad Real and Jaén). Overall, we found 33 polymorphic sites, 20 of which were parsimoniously informative. In the coding region, 30 polymorphic sites were found, 18 were parsimoniously informative and 15 coded for a different amino acid. Eight changes occurred in the first codon position, 7 in the second and 15 in the third. The highest frequency of nucleotide changes was found at the terminal regions of the MV genome, mainly between nucleotides 146,634 and 146,842, corresponding to genes 150R and 151R (Fig. 2).

The 15 amino acid substitutions revealed 19 viral proteotypes. Five proteotypes encompassed two or more nucleotide haplotypes (Table 3). At the amino acid level, we found that five haplotypes (MV18–20, MV32 and MV35) presented the same proteotype as the Lausanne strain (MV_Aa1). Furthermore, MV_Aa1 was the second most common proteotype, after MV_Aa2, which contained five haplotypes, including the most common one, MV2. The proteotype that had the largest number of amino acid changes compared to the Lausanne strain was MV_Aa5, with five amino acid substitutions (Table 3).

All of the coalescence models analysed in BEAST 1.4.6 based on a relaxed molecular clock had a better fit than the models that were forced to fit with a strict molecular clock. Overall, the best model for our data was the relaxed clock and constant population size (Table 4). Considering this model, we estimated a mutation rate of 6.26×10^{-5} substitutions per site and per year (95% HPD: 3.69×10^{-5} – 9.00×10^{-5}).

We observed a modal mismatch distribution for the whole of the data set and coding region (Fig. 3), with low non-significant Hri values ($Hri = 0.042$ and 0.044 , respectively), and thus did not reject the null hypothesis of constant population size. The non-coding regions showed a very low number of inter-haplotype pair nucleotide changes (Fig. 3) and a high, although non-significant, Hri value ($Hri = 0.808$; Table 5), which indicates that this data set does not adjust to a population expansion model.

All of the neutrality tests performed were significantly negative for the complete data set and coding regions, which indicated a surplus of low-frequency haplotypes and a consequent deviation from neutrality. However, for the non-coding regions, only Tajima's *D* and Fu and Li's *F** were significant (Table 5).

No recombination points were detected in the coding region data set, and therefore all of these sequences were analysed together to identify positively selected sites. In addition, we did not find any strain that had a significant proportion of different ancestors according to BAPS 4.14, which could also be indicative of recombination [13]. The mean d_N/d_S value was 0.563 (95% CI = 0.382–0.793). The most conservative methods, SLAC and FEL, detected one positively selected site, (codon 625 of the coding regions

Fig. 3 Mismatch distribution of the alignments constructed for the entire data set, the coding regions and non-coding regions of the analysed MV fragments. Grey bars represent the observed distribution, solid lines the expected distribution under a growing or declining population model, and the dotted line the expected distribution under a constant population size model

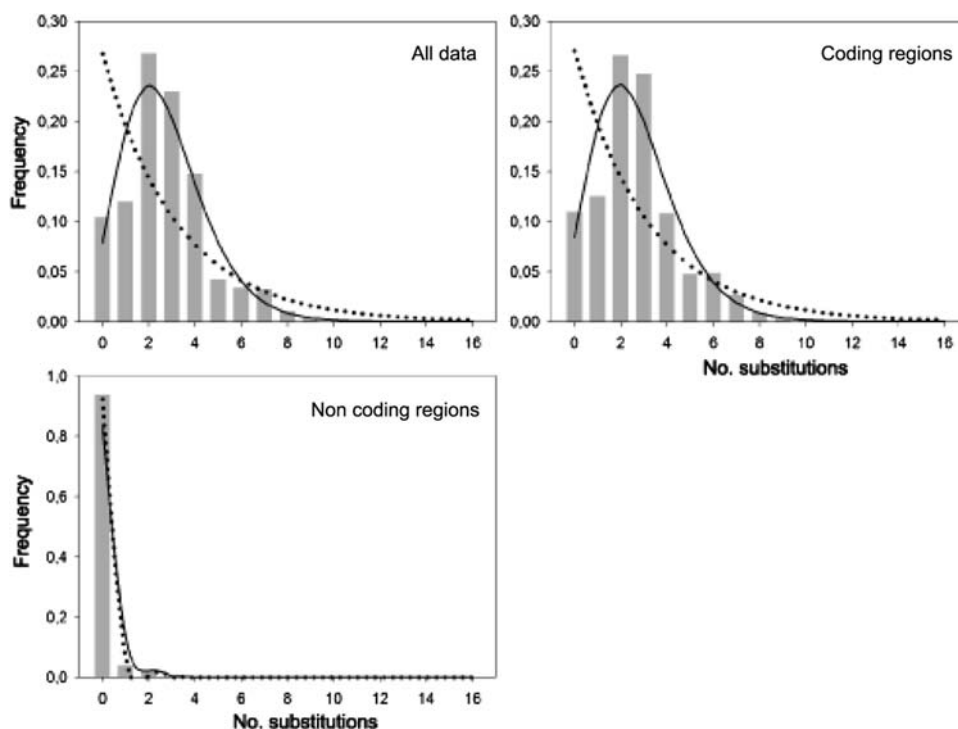


Table 5 Neutrality tests and demographic parameters estimated for all the MV data, the coding regions and the non-coding regions

	All	Coding	Non-coding
Demographic parameters			
H _{ri}	0.041	0.044	0.811
θ_0	0.731	0.786	0.19
θ_1	1000	1000	1000
τ	2.007	1.911	0
Neutrality tests			
Tajima's <i>D</i>	-1.761*	-1.636*	-1.549*
Fu's <i>F</i>	-31.756**	-28.218**	-2719
Fu and Li's <i>D</i> *	-2.3156*	-2.022*	-1991
Fu and Li's <i>F</i> *	-2.514*	-2.241*	-2.172*
<i>R</i> ₂	0.039**	0.043*	0.067

* Significant at $P < 0.05$

** Significant at $P < 0.01$

The NJ tree did not show high bootstrap values that supported the phylogenetic relationships between MV haplotypes. The Lausanne strain haplotype was placed at a basal position with respect to the current MV strains, but without any bootstrap support (Fig. 4). The haplotype network showed multiple connections between haplotypes and lacked a clear genetic structure. The most common haplotype, MV2, was the central haplotype, whereas the Lausanne haplotype was found at a terminal position of the network, connected to another four haplotypes (MV3, MV32, MV19 and MV35; Fig. 1).

The Bayesian clustering method detected the presence of a genetic structure for the MV samples analysed. The best structure showed three genetic groups: A, B and C (Fig. 1), with a posterior probability greater than 0.99. Genetic group B was the most frequent, containing 58 samples, group A contained 22 samples, including the Lausanne strain, and group C contained 18 samples (Fig. 1). Group C showed the widest geographic distribution, being found in Mallorca, Galicia, Ciudad Real, Albacete and Jaén. Group A had the most restricted distribution, occurring only in Toledo1 and 3, Ciudad Real and Jaén. In three of these locations, we also found group C; however, viruses from group A, which were detected in five localities, were never found in sympatry with viruses from any of the other genetic groups (Fig. 1).

Discussion

Myxoma virus has been the subject of numerous studies dealing with its infection and virulence mechanisms [20, 21, 71] as well as with its relationship to the resistance of rabbits to myxomatosis [7, 8, 35, 36]. However, molecular studies from field strains are scarce and limited to local studies whose objective was to characterise viral strains using poorly informative markers, such as RFLPs [41, 55, 62]. Also, most of these studies were carried out in areas where the rabbit is considered to be a pest, such as Australia, North America and certain areas of Europe.

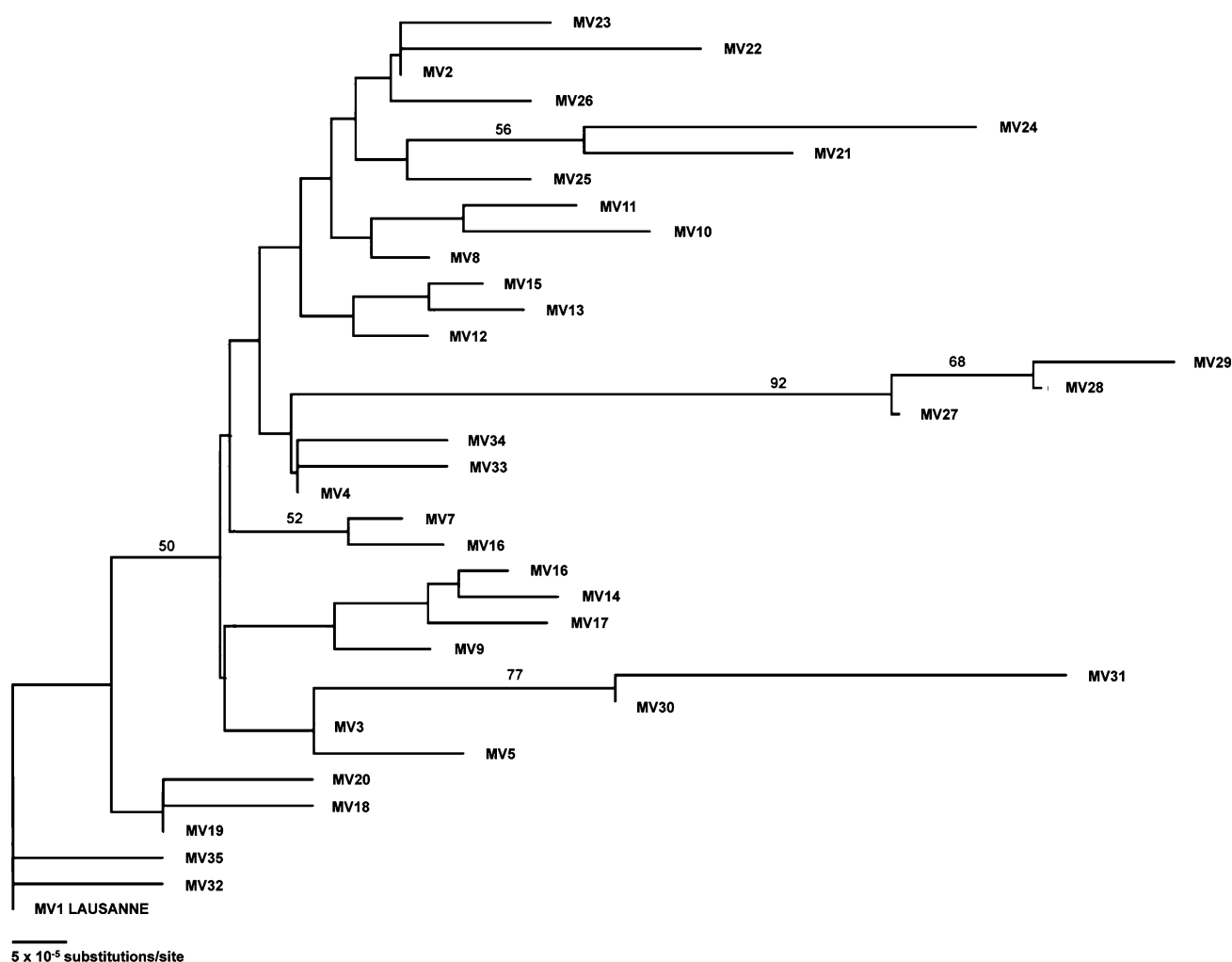


Fig. 4 Neighbour-joining phylogenetic tree inferred for the MV haplotypes found in Spain. Bootstrap values greater than or equal to 50 are shown

In Spain, among the 7,681 characters analysed, only 33 polymorphic sites were detected ($\pi = 0.00036 \pm 0.00003$), which gave 35 haplotypes of MV. The average genetic distance between haplotypes ($D_P = 0.00056 \pm 0.00012$) was within the range described for other poxviruses and similar to that of the genus *Capripoxvirus* [46]. Despite their wide range of hosts, the poxvirus genera show highly conserved genomes. Even though some studies have observed large divergences among strains of the same virus, this is not the general rule [46]. In our case, we did find genetic changes from the original introduced strain, but the genetic variability at the nucleotide and amino acid level was lower than that observed for other poxviruses in mammals and birds [40, 67, 69]. This low variation is not surprising, however, as it corresponds to that predicted for large double-stranded DNA viruses [15, 19, 32].

The distribution of genetic variability was not uniform across the analysed regions of the MV genome. Most of the polymorphic sites were located in the terminal regions of

the MV genome, mainly at the right end (Fig. 2). This pattern is common for all of the poxviruses infecting chordates (subfamily *Chordopoxvirinae*) that share 90 highly conserved genes located in the central 100 kb of the genome. None of the genes found at the terminal regions of the genome are shared by all members of this subfamily [29].

Four of the fragments analysed in this study belong to the central portion of conserved genes (M029–M031, M068–M071, M077–M079 and M089–M091) and contain most of the structural and housekeeping genes. The terminal regions, mainly M011–M013 and M150–M152, code for proteins that either regulate apoptosis, inhibit the synthesis of interferon, or have an anti-inflammatory function [9]. These functions are closely associated with the strategy of the virus to modulate the host's immune response [50, 68, 71]. Moreover, the genetic changes detected in these regions could influence the virulence of these strains. As has been observed, in the highly virulent MSW strain

M151R and M152R, genes are duplicated [41], whereas when the strains are knocked out for M150R, M151R and M152R, the virus is attenuated [10, 30, 47].

It has been proposed that the complex mechanisms that allow poxviruses to evade the host immune response may have played an important role in the evolution of these viruses and would have been under different selective pressures during the co-adaptation of rabbits and MV [71]. This is supported by the significant deviations from neutrality that are shown by all of the tests that were carried out for the complete data set and coding regions. Although these deviations could also result from population size changes, other evidence indicates that they are more likely to be due to natural selection. Firstly, the best demographic model for our data, according to the calculated BF, was a constant population size model. Furthermore, this method in all cases indicated that the data did not fit with a strict molecular clock, which indicates that the mutation rate varies along the branches of the tree. This could also be the result of selective pressures on certain viral strains. However, for the non-coding regions, only Tajima's D and Fu and Li's F^* were significant, and R_2 , which is the most appropriate test for small sampling sizes and low segregating site numbers [57], was not significant (Table 5). Even the power of the analysis could be reduced by the low polymorphism of the data; these results seem to indicate that the non-coding regions would better fit a neutral evolutionary model, as they would not be under a strong selective pressure. Also, the constant effective population size of the virus could indicate that the genetic diversity has not increased significantly since the introduction of MV in 1953 [49]. This supports the low divergence observed in the current MV field and Lausanne strains and the fact that although the virulence of MV in Spain initially decreased, today it is still high [6].

On the other hand, the average ratio of non-synonymous to synonymous substitutions indicates that the coding regions are probably under some kind of purifying selection ($0 < d_N/d_S < 1$; [37]) that favours the appearance of intermediate phenotypes, as the moderately virulent strains may be [25, 44], and that in the long-term would contribute to a decrease in genetic diversity [59]. The selective pressure over the codons was not the same for all of the codons across the MV genome. In the central region, most of the codon substitutions were negatively selected, except for codon 625 in M070R, which was detected by all methods as being under positive selection. This agrees with the fact that these are structural genes, which are essential for the virus viability. Therefore, any mutation suffered by these genes would probably be negatively selected [29, 42]. On the other hand, beyond 123 kb at the right end, we found a high density of both positively and negatively selected sites (Fig. 2), which indicates that the virulence

genes located in this region are under a strong selective pressure and could be involved in the above-mentioned equilibrium between virulent and attenuated strains. Above all, we must consider that the conservative methods used to analyse evidence of selection (SLAC and FEL), only detected one positively selected (66,697–66,699 M070R) and two negatively selected (14,423–14,425 M012L and 76,345–76,347 M079R) sites. However, most of the other codons under some kind of selection were detected by REL, which is likely to produce type I errors with small data sets. Therefore, considering the low polymorphism of these data, we consider these results cautiously.

The type of selection is a key factor, as well as the viral genetic material (RNA, DNA, single- or double-stranded), genome size, generation time, existence of viral enzymes for DNA reparation, or type of transmission (direct or vector-borne) to determine the evolutionary rate of a virus [19]. Considering these factors as well as genetic and epidemiological characteristics of MV, we could explain the low estimated substitution rate for MV in this study (6.26×10^{-5} substitutions/site/year). This rate, however, was much higher than the mutation rate estimated for both the single- and double-stranded DNA virus (3.5×10^{-8} – 7.2×10^{-7}) [16, 63] and the other poxviruses (9×10^{-7} – 1.2×10^{-6}) [4].

The differences observed in the substitution rates might be the result of the different methods used to estimate them. Firstly, the low evolutionary rate obtained in previous studies could be due to the use of RFLPs or small conserved gene fragments, which had a much lower genetic variability [41, 55, 62]. Secondly, previous studies have used strict molecular clocks that were calibrated according to the divergence times of the hosts [4, 63], instead of applying coalescent-theory-based methods. The MCMC allows the incorporation, for each taxon, of temporal information into the phylogenetic information and of different substitution rates among lineages [17]. However, these methods assume that all sampled sequences contain only fixed substitutions and do not take into consideration that some of these mutations might be transient and may not be inherited by the next generations. Therefore, the estimated evolutionary rates reflect the mutation and/or substitution rates. The latter will consequently lead to an overestimation of the substitution rate when the sequences have been sampled during a short period of time, because this rate will be more similar to the mutation rate [19].

In addition to the methodological aspects, there are other intrinsic features of MV that could make it have different evolutionary rates with respect to other viruses or geographical regions. Unlike other poxviruses, MV is only virulent in a single species, which is not its natural host. MV has thus not evolved in the rabbit genus *Oryctolagus*, but in *Sylvilagus*, in which it causes localised and benign

infections [22]. Therefore, the differences present in the *O. cuniculus* immune system that make them susceptible to infection [1], and consequently, their immune response to the infection, would have represented a new selective pressure for MV [35]. In this regard, the rapid adaptation of MV to areas where it had been introduced and the proliferation of attenuated strains [43, 45, 70] could explain the increase in the number of fixed mutations compared to that of other poxviruses that could already be in equilibrium with their natural hosts because of an older existing relationship between the two.

In addition, the great dispersion experienced by MV since its introduction has caused the virus to occupy a wide variety of habitats, which would eventually be conditioned by the presence and abundance of arthropods, the main vectors of myxomatosis [22]. Therefore, these factors associated with MV's vectors might have influenced the evolution of the virus in the different geographic regions: the reproductive biology of the vector and its abundance and diversity can determine the epidemiology of MV [25, 52].

In contrast to RNA viruses, which can provide genetic information on their spatial and temporal dynamics on a smaller scale, the phylogeographical study of DNA viruses is more restricted due to their lower evolutionary rates. This agrees with the absence of a phylogenetic structure detected among the MV strains analysed (Fig. 4) and the low or null differentiation found using RFLPs within a single population, where most of the variation was found at the ITRs [41, 55, 61, 62]. Also, no correlation was found between MV and the strong genetic structure at nuclear and mitochondrial markers in rabbits of the Iberian Peninsula [2], which could indicate that MV has the same capacity to infect rabbits, independent of their evolutionary lineage. Hence, it appears that the short time period since the introduction of MV to Spain in 1953, the low mutation rate and the large virus population size [32] have not still permitted, solely by drift, the divergence of MV lineages or the existence of any geographical differentiation. Any existing differentiation may also be disrupted by the continuous movement of rabbits that has gone on in Spain during the many translocations [14].

The genetic differences between strains have allowed the determination of the origin of the virus introduced in certain regions of Australia [62]. In the MV phylogenetic analysis carried out in Spain, the Lausanne strain, from which, theoretically, all current strains stem [49], appeared at the base of the tree (at a branch with a length of zero), which could be interpreted as an ancestral haplotype [54]. However, when a more appropriate genealogical reconstruction was applied in which it is not assumed that haplotypes must be located at the tip of the branches, the

Lausanne strain was neither in a central network position nor was it the most common haplotype, as would be expected for an ancestral haplotype in a neutral coalescent model [11].

If we consider that selection has an important effect on the evolution of MV, as is shown by the neutrality and selection tests performed as well as by the rejection of a strict molecular clock, the Lausanne strain could be regarded as being the real ancestral strain of the current MV field strains in Spain. There could have also been a positive selection for less virulent strains, such as the strains from group B. The new positively selected strains would increase their frequency in the population and make up the central haplotypes within each of the inferred genetic groups: MV19, has the same amino acid sequence as Lausanne, in group A; MV2 in group B and MV5 or MV6 in group C. From these, other derived haplotypes could be better or worse adapted than their ancestors (Fig. 1). However, the difficulty in finding infected animals in the wild did not permit us to have even samples, and therefore further efforts should be made to investigate these issues. The inferred genetic groups correspond almost perfectly to the virus proteotypes (Table 3), and although they do not show a clear geographical structure, their frequencies do differ in the different locations, so local adaptations could possibly be occurring.

In the future, it will be necessary to study the geographic distribution and the virulence of the different MV genetic groups and investigate if they are related to some of the amino acid changes detected. Thus, it would be possible to determine if the presence of certain strains is associated with a higher incidence of myxomatosis in different rabbit populations, and consequently manage and preserve them more appropriately.

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